

SMART DINE: GRAND RESERVE

Advanced AI-Powered Full-Stack Restaurant Management & Operations Platform

Internship Project Final Technical & Operational Report

Project Metadata & Attribution

Attribute	Details
Project Title	Smart Dine (Grand Reserve) - Unified Restaurant Operations System
Developer / Intern	Massna Ijaz
Internship Mentor	Sir Aznair
Live Platform	Smart Dine Cloud Production
Source Repository	Smart-Dine GitHub Repository

Default System Demo Credentials

Role / Station	Account Identifier	Access Key	Primary Function
Manager Console	manager@smartdine.pk	12345678	Executive analytics, P&L, staffing, and AI settings
Waiter Station	wa@gmail.com	12345678	Dynamic floor mapping, instant ordering, and status tracking
Kitchen KDS	ki@gmail.com	12345678	Real-time prep dispatch, ticket routing, and metrics

Role / Station	Account Identifier	Access Key	Primary Function
Cashier Billing	ca@gmail.com	12345678	Order settlement, discounts, auto-change, and QR receipt generation

1. Executive Summary

Smart Dine: Grand Reserve is an enterprise-grade, real-time restaurant management ecosystem designed to eliminate operational bottlenecks in fine-dining and high-volume dining establishments. Built from the ground up to solve communication delays between floor staff, kitchen operations, billing counters, and executive decision-makers, Smart Dine provides four dedicated role-specific portals synchronized via live state pipelines.

In addition to traditional POS and ticketing capabilities, Smart Dine integrates predictive artificial intelligence modules:

- **Predictive Demand & Kitchen Planning:** Machine learning forecasting that analyzes historical sales to predict future portion demands and prevent food stockouts or over-preparation.
- **AI Executive Assistant & Voice Operations:** A conversational LLM and real-time voice interface allowing managers to query complex financial metrics, inventory states, and item contributions hands-free.
- **Automated Recommendation Engine:** Algorithmic detection of slow-moving inventory, high-margin combo promotions, and automated reorder replenishment alerts.
- **Receipt-Verified QR Feedback Loop:** A cryptographic, single-use QR review system embedded directly on customer payment receipts to prevent fraudulent reviews and feed structured customer sentiment back to executive analytics.

2. System Architecture & Role-Based Workspaces

The platform operates on a single unified database with four specialized user portals:

A. Executive Service Console (Manager Portal)

The managerial command center provides complete organizational control and real-time business intelligence:

- **Operations Dashboard:** Live counts of active staff, menu offerings, ready tickets, and total orders served/paid (450+ orders tracked).
- **15-Day Sales Outlook:** Statistical weekday-weighted sales forecast projecting expected customers (e.g., 18 guests/day) and projected daily revenue (Rs 19,150+ average, with busy-day peaks up to Rs 23,000).
- **Comprehensive P&L Statement:** Real-time financial accounting breaking down Net Revenue, Cost of Goods Sold (COGS based on recipe ingredients), Gross Margin (e.g., 82.4%), Operating Expenses (OpEx), and Net Profit.
- **Demand Planning:** 6-month historical training sets validating predictive accuracy across 12-week cycles to output dish-level preparation forecasts.
- **Operating Expense Tracking:** Modular expense ledger categorizing raw ingredients, rent, salaries, utilities, maintenance, and marketing with status validation.
- **Floor & Table Management:** Dynamic floor builder supporting multi-level floor plans, custom seating capacity, and table-level tax configuration.
- **Staff Administration:** Full credential management, station assignment, password reset, and status deactivation controls.

B. Dining Room Service (Waiter Station)

Designed for mobile-first floor efficiency:

- **Rapid Order Entry:** Interactive floor and table selector, optional kitchen preparation notes (e.g., "extra spicy please"), category filters (Kitchen Dishes vs. Inventory Drinks), and instant subtotal/tax computation.
- **Service Station Flow Tracker:** Real-time visibility into tickets transitioning across Pending, Preparing, Ready, and Completed stages.

- **Performance Analytics:** Personal waiter metrics tracking active tables, 30-day settled sales (Rs 497,000+), orders served (95+), and guests accommodated.

C. Kitchen Culinary Dispatch (Kitchen Display System - KDS)

Replaces paper tickets with a synchronized culinary workflow:

- **Tri-Stage Order Pipeline:**
 - a. *Pending Dispatch:* Newly created orders awaiting kitchen acknowledgment (Start Preparation).
 - b. *In Preparation:* Active cooking stage, automatically deducting recipe inventory ingredients (Ready for Pickup).
 - c. *Ready & Plated:* Staged for server pickup and marked as Awaiting Payment.
- **Culinary Metrics:** 30-day ticket throughput tracker (105+ tickets, 565 units prepared) and station-handling distribution.

D. Cashier & Billing Station

High-reliability billing terminal:

- **Live Unpaid Queue:** Instant retrieval of completed floor tickets ready for settlement.
 - **Flexible Discount Engine:** One-click percentage discounts (5%, 10%, 15%, 20%) or custom rates with mandatory audit reasons.
 - **Cash Management & Auto Change Calculator:** Real-time computation of cash received versus change due to eliminate manual cashier math errors.
 - **Thermal Receipt Generation with Dynamic QR:** Direct print support for receipts embedding one-time customer review QR codes.
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3. Advanced Artificial Intelligence & Data Intelligence Modules

3.1 Smart Dine Assistant (Text & Voice AI)

- Built directly into the manager console via *Ask Smart Dine* modal and *Smart Dine Live* voice interface.
- Capable of complex analytical queries:
 - **Query:** "Which menu items have the highest profit margin and contribute most to net income?"
 - **Output:** Breaks down beverage margins (Fanta/Coke at 86%+) versus net income volume drivers (Beef Plao0 generating Rs 497k, Chicken Saji generating Rs 330k).
- Real-time voice interaction enables managers to perform audit inquiries on the move.

3.2 Automated Recommendation Engine

- **Strategy Categories:** Combo & Promotion Optimization, Stock Replenishment, and Margin Preservation.
- **Audit Workflow:** Generates actionable proposals with predicted business impact (e.g., promotional pairing for Pepsi 500ml to boost average ticket size, or reorder warnings when CocaCola stock drops below minimum threshold).
- **Manager Approval Workflow:** Managers can review, approve, apply live change, or reject suggestions.

3.3 Receipt-Linked Customer Review Engine

- Solves review fraud by generating a unique, receipt-anchored QR code upon payment.
 - **One review per transaction token:** Once submitted, the token is invalidated.
 - Mobile survey captures 5-star overall rating, individual dish-level ratings (Chicken Saji, Biryani, etc.), and 5 operational criteria (Taste, Speed, Cleanliness, Hospitality, Value).
 - Immediate ingestion into the Manager's *Guest Reviews & Service Quality* dashboard with automated sentiment tagging.
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4. End-to-End Operational Lifecycle Walkthrough

To illustrate the seamless integration, the system handles an end-to-end transaction through the following six steps:[1. Order Capture] —> [2. Kitchen Dispatch] —> [3. Culinary Pass]

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[6. Audit Log] <— [5. Customer Review] <— [4. Cashier Settlement]

1. **Order Capture:** Waiter selects Floor 1, Table 2, adding Chicken Saji, Chicken Biryani, Beef Plao, Fanta, and Pepsi with special instruction *"spicy please"*. Order ID **SD-100003** is created in **PENDING** state.
2. **Kitchen Dispatch:** Ticket immediately appears on Ali's Kitchen KDS screen. Ali clicks *Start Preparation* (status updates to **PREPARING** across all portals).
3. **Culinary Pass:** Upon meal completion, Ali marks *Ready for Pickup*. Waiter is notified to deliver dishes to Table 2.
4. **Billing Settlement:** MOOn (Cashier) opens ticket **SD-100003** for Rs 4,960. Customer tenders Rs 5,000 cash. The system calculates Rs 40 change and stamps the bill as **PAID**.
5. **Customer Feedback:** The printed bill features a dynamic QR code. The customer scans it on mobile, rates the meal 5 stars (*"Very good"*), and submits.
6. **Audit & Attribution:** The completed order is permanently logged with complete multi-role accountability (Waiter: **raza**, Chef: **ali**, Cashier: **MOOn**) and reflected in executive P&L and review metrics.

5. Technology Stack & Key Highlights

- **Frontend Architecture:** Modern, responsive component-based UI with luxury dark-emerald styling, real-time live synchronization (15-second polling/WebSocket state updates), and mobile-responsive viewport.
- **Backend & API:** RESTful operational endpoints, role-based authentication guard, and live transactional state management.
- **Analytics & Forecasting:** Time-series demand forecasting algorithm using sliding 6-month historical windows with 12-week validation sets.

- **AI Integration:** LLM operational analytics agent with natural language query parsing and WebRTC/Voice interactive modal.
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6. Conclusion & Acknowledgments

The **Smart Dine** platform demonstrates a comprehensive, full-stack software engineering achievement, effectively combining real-time multi-terminal operations with cutting-edge artificial intelligence. Sincere gratitude is extended to **Sir Aznair** for mentorship, architectural guidance, and leadership throughout this project development lifecycle.

